

TEMPLATE PURCHASE DESCRIPTION

INSTRUCTIONS: This template should be used when purchase equipment, products, supplies, software, software license agreements, warranty agreements, and maintenance service agreements. Remove the instructions from the purchase description before submitting with the purchase request.

PURCHASE DESCRIPTION

1. **Statement of Need and Purpose:** Investigators in the HANDLS study require data on circulating humanin (MT-RNR2) levels to support an ongoing longitudinal study. This data will expand our current research on examining how circulating factors play a role in age-related chronic disease. Humanin has anti-apoptotic, metabolic and neuroprotective properties and also has been associated with lifespan. It is important to quantitatively measure serum levels of humanin in HANDLS participants to understand the role of this hormone in aging, and age-related chronic disease, including type 2 diabetes. The government requires specific ELISA kits because the government has specific expertise, equipment and resources to perform the work using these kits at our Institute.
2. **Background Information and Objective:** Humanin (MT-RNR2) is a small, mitochondria-derived peptide that has attracted significant scientific interest due to its cytoprotective and anti-apoptotic properties. Humanin has been associated with aging and age-related chronic diseases including type 2 diabetes, neurodegenerative diseases and cardiovascular disease. To gain insight into its role in these processes in human cohorts, we will quantitatively measure levels of circulating humanin using ELISA kits. We will measure human humanin from serum from HANDLS participants. Serum samples from HANDLS participants are precious and limited quantities are available. Therefore, only a limited amount (<50 ul) can be used for this ongoing longitudinal study. We previously measured humanin using an ELISA kit (POTS# 24-012790) in a pilot study of 176 HANDLS participants. The government requires additional kits to expand upon this initial pilot study and to support an ongoing longitudinal study. For continuity of research and comparability of results, we need to use the same kit (CSB-EL015084HU) to measure humanin in additional HANDLS participants.
3. **Generic Name of Product:** Human Humanin ELISA Kits
4. **Purchase Description:**

Human putative humanin peptide (MT-RNR2) ELISA kits
5. **Salient characteristics**
Human Humanin ELISA kit
 - ELISA technology
 - 96 well format
 - one kit assays humanin in 96 samples (including standards)
 - measure humanin from human serum
 - sensitivity to measure humanin at <7 pg/ml
 - required volume of serum to be <50 ul
 - compatibility to read absorbance on a SpectraMax M2 Microplate Reader
 - for continuity of research we need to compare our new data to our previous results using the

Human Putative humanin peptide (MT-RNR2) ELISA kits (CSB-EL015084HU).

-for the validity of the scientific research, data generated will need to be directly compared to those previously published in human aging cohorts. The human putative humanin peptide (MT-RNR2) (MT-RNR2) ELISA kits (CSB-EL015084HU) was used to assay humanin in human aging cohorts. Therefore for the validity and direct comparability of our results to our previous results and those previously published, the government requires this same kit to assay the levels of humanin.

- 6. **Quantity:** 49 Kits
- 7. **Delivery Date:** 2 Weeks ARO
- 8. **Period of Performance:** N/A
- 9. **Option(s):** N/A